

Temperature-induced degradation of GaN HEMT: An *in situ* heating study



Cite as: J. Vac. Sci. Technol. B 42, 032209 (2024); doi: 10.1116/6.0003490

Submitted: 17 March 2024 · Accepted: 11 April 2024 ·

Published Online: 26 April 2024



View Online



Export Citation



CrossMark

Md Abu Jafar Rasel,¹ Di Zhang,² Aiping Chen,² Melonie Thomas,¹ Stephen D. House,³ Winson Kuo,² John Watt,² Ahmad Islam,⁴ Nicholas Glavin,⁵ M. Smyth,³ Aman Haque,^{1,a)} Douglas E. Wolfe,⁶ and Stephen J. Pearton⁷

AFFILIATIONS

¹Department of Mechanical Engineering, Pennsylvania State University, University Park, Pennsylvania 16802

²Center for Integrated Nanotechnologies (CINT), Los Alamos National Laboratory, New Mexico 87545

³Center for Integrated Nanotechnologies (CINT), Sandia National Laboratories, New Mexico 87185

⁴Sensors Directorate, Air Force Research Laboratory, Wright-Patterson AFB, Ohio 45433

⁵Materials and Manufacturing Directorate, Air Force Research Laboratory, Wright-Patterson AFB, Ohio 45433

⁶Department of Materials Science and Engineering, Pennsylvania State University, University Park, Pennsylvania 16802

⁷Department of Material Science and Engineering, University of Florida, Gainesville, Florida 32611

^{a)}Author to whom correspondence should be addressed: mah37@psu.edu

ABSTRACT

High-power electronics, such as GaN high electron mobility transistors (HEMTs), are expected to perform reliably in high-temperature conditions. This study aims to gain an understanding of the microscopic origin of both material and device vulnerabilities to high temperatures by real-time monitoring of the onset of structural degradation under varying temperature conditions. This is achieved by operating GaN HEMT devices *in situ* inside a transmission electron microscope (TEM). Electron-transparent specimens are prepared from a bulk device and heated up to 800 °C. High-resolution TEM (HRTEM), scanning TEM (STEM), energy-dispersive x-ray spectroscopy (EDS), and geometric phase analysis (GPA) are performed to evaluate crystal quality, material diffusion, and strain propagation in the sample before and after heating. Gate contact area reduction is visible from 470 °C accompanied by Ni/Au intermixing near the gate/AlGaIn interface. Elevated temperatures induce significant out-of-plane lattice expansion at the SiN_x/GaN/AlGaIn interface, as revealed by geometry-phase GPA strain maps, while in-plane strains remain relatively consistent. Exposure to temperatures exceeding 500 °C leads to almost two orders of magnitude increase in leakage current in bulk devices in this study, which complements the results from our TEM experiment. The findings of this study offer real-time visual insights into identifying the initial location of degradation and highlight the impact of temperature on the bulk device's structure, electrical properties, and material degradation.

Published under an exclusive license by the AVS. <https://doi.org/10.1116/6.0003490>

I. INTRODUCTION

Wide-bandgap devices, such as AlGaIn-GaN high electron mobility transistors (HEMTs),^{1–14} have become increasingly popular because of several advantages over traditional silicon-based transistors in extreme environments, such as higher bandgap, lower intrinsic carrier concentration, excellent power handling capabilities, etc.¹⁵ Next-generation radar systems, satellite communications, and high-frequency wireless communication devices require microwave power amplification and high-power, high-temperature applications. Temperature affects the GaN-based devices, such as UV

sensors as well as GaN HEMT.^{16–18} GaN HEMTs are used in harsh environments where traditional silicon-based transistors may not operate efficiently, making them suitable for downhole drilling, geothermal exploration, automotive for efficient power conversion, and space missions. The thermal stability of these structures and changes in stress related to thermal excursions are keys for stable operation.^{11,12} Apart from applications in high-temperature environments, AlGaIn/GaN HEMTs designed for room-temperature, high-power applications may experience device self-heating. This self-heating phenomenon can elevate the channel temperature